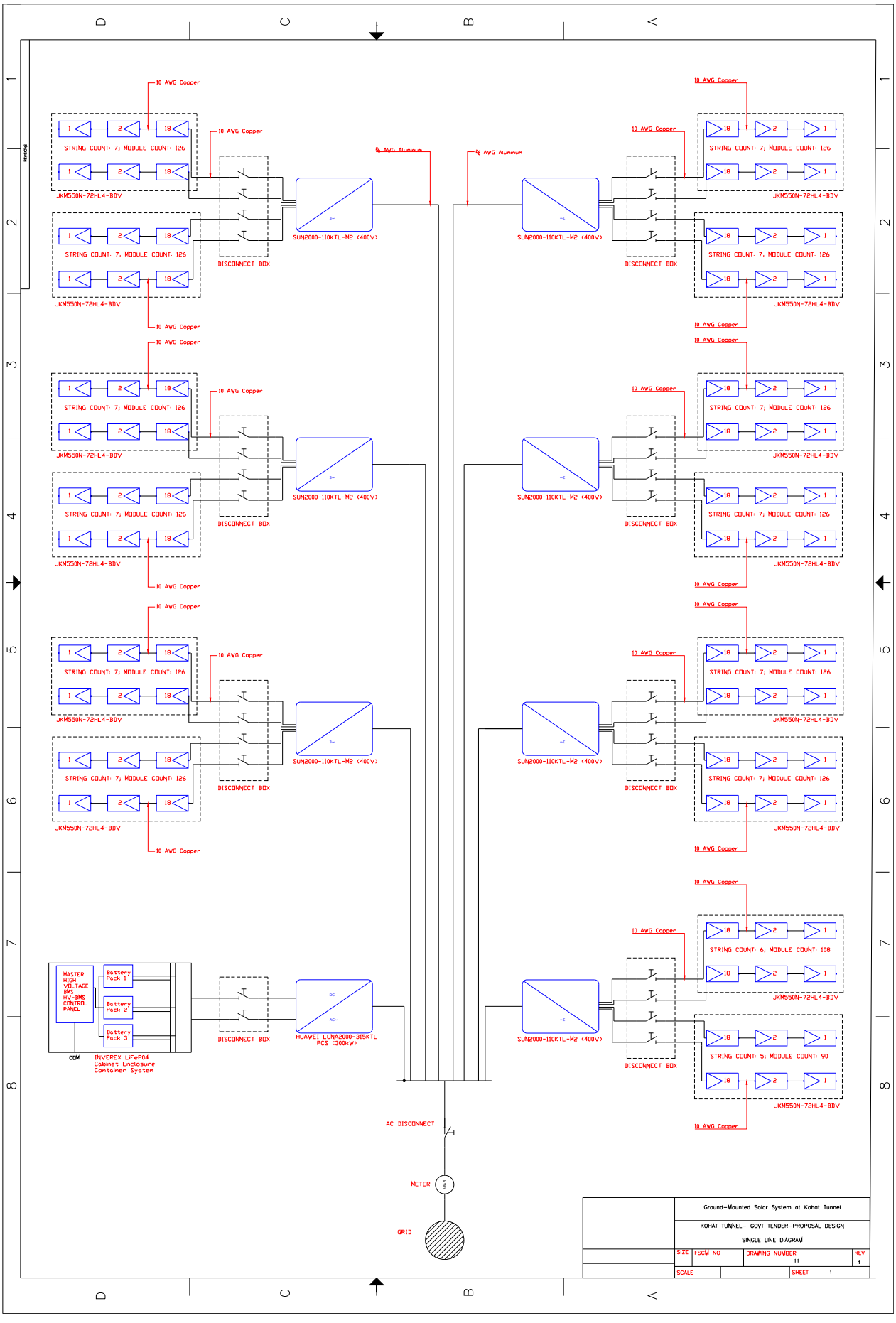
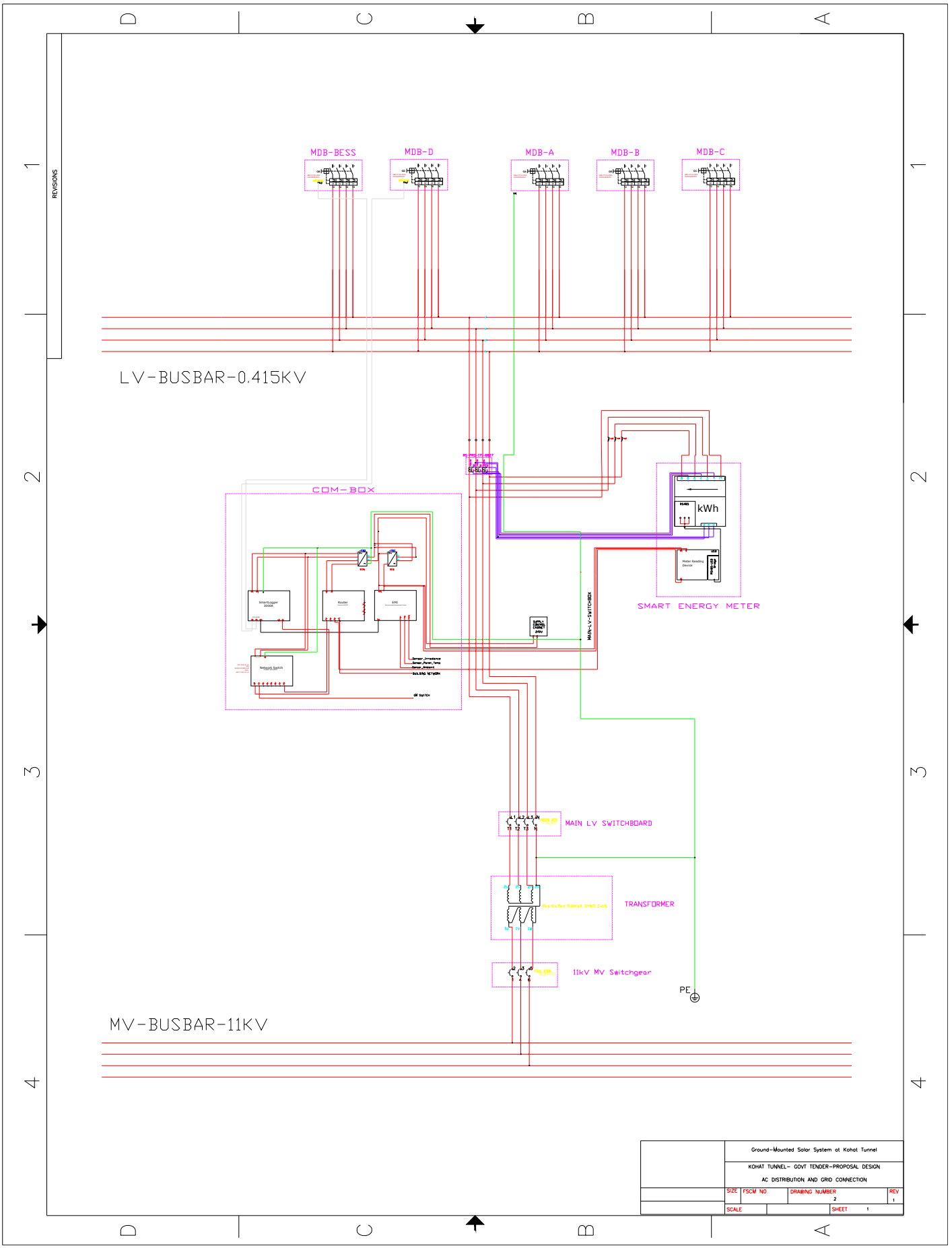
PROJECT: KOHAT TUNNEL

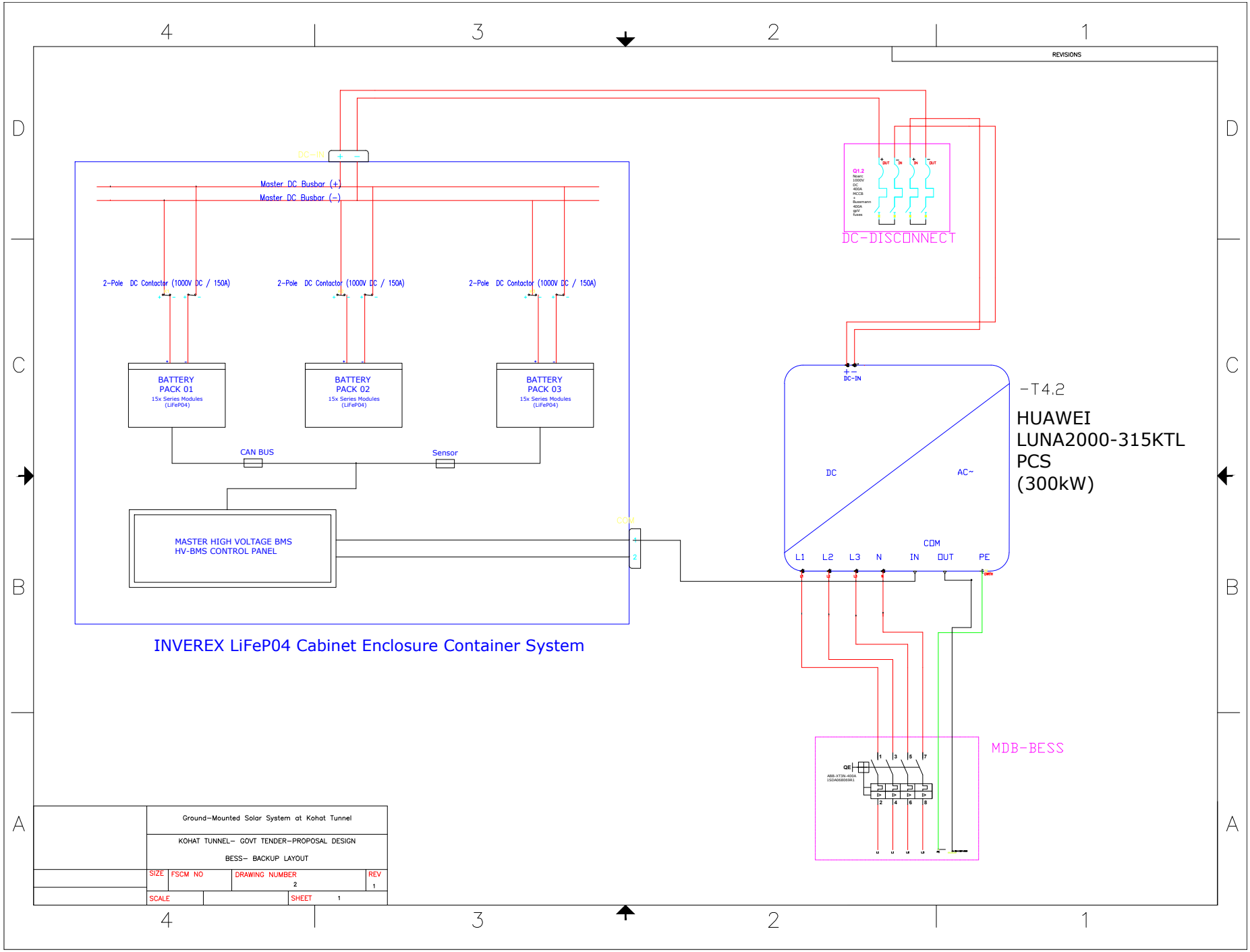
Solar Energy System Wiring Diagram

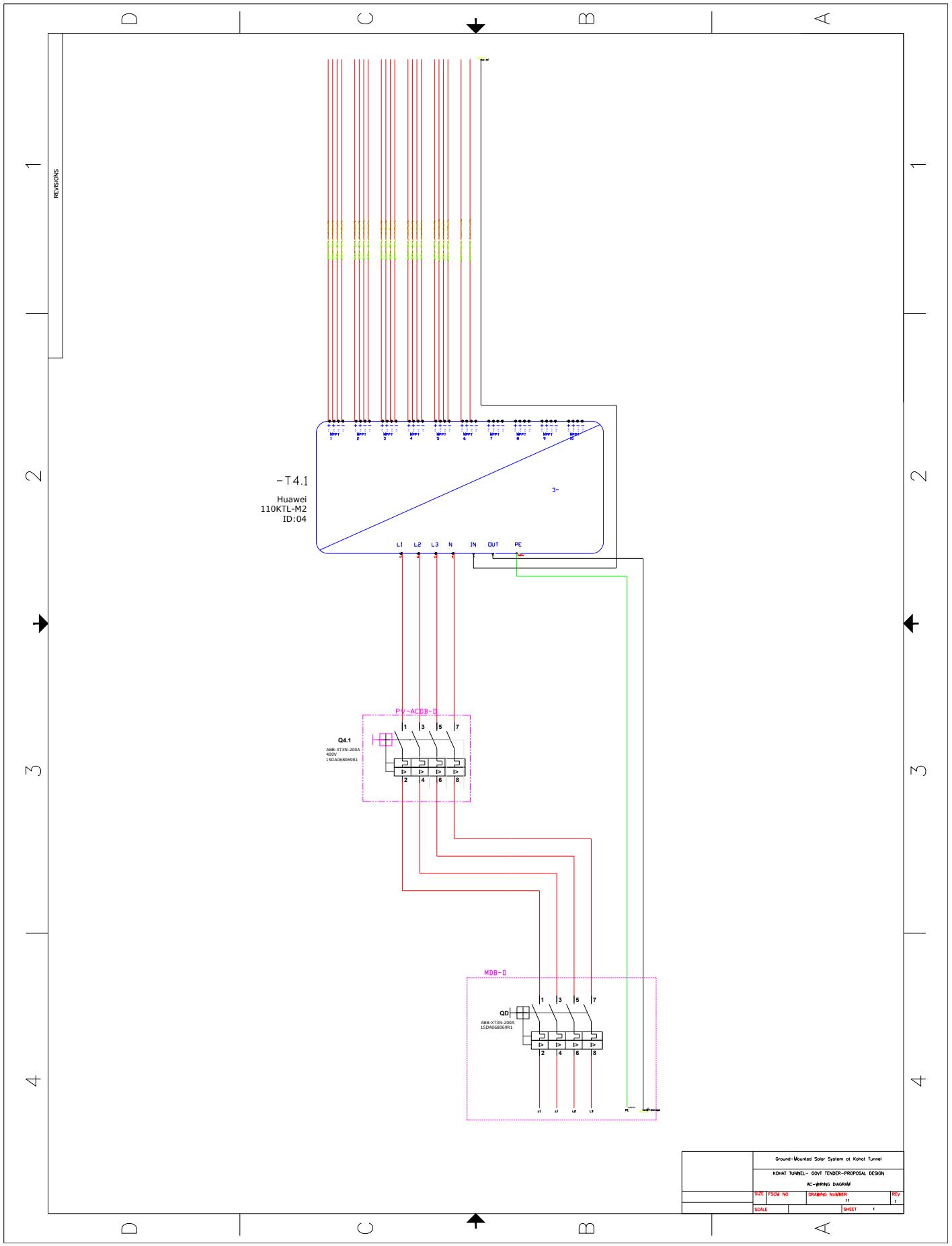
770kW Solar Energy System with 863.4kWp PV and 600kWh Battery Energy Storage System

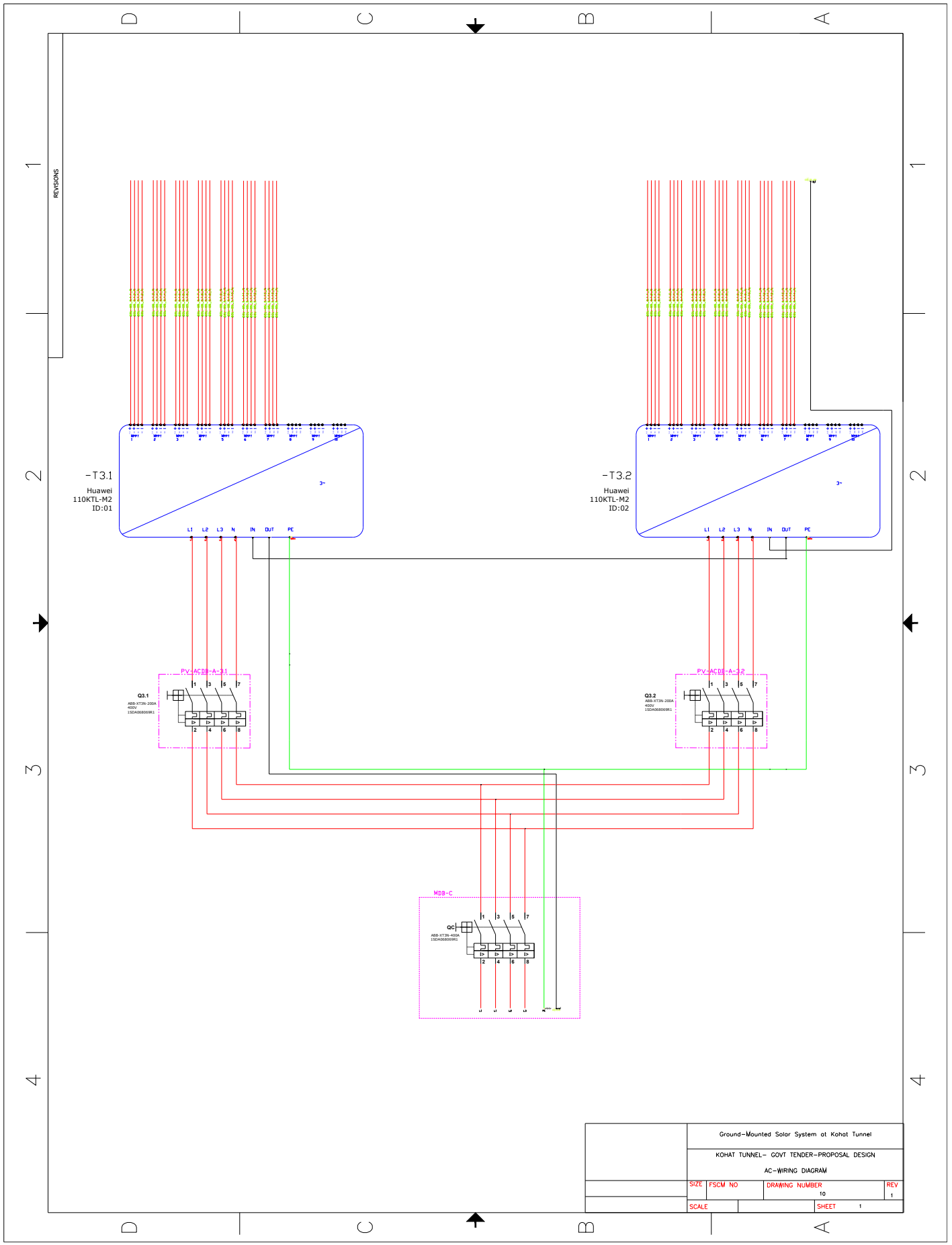
Engr. M Usama Rafique

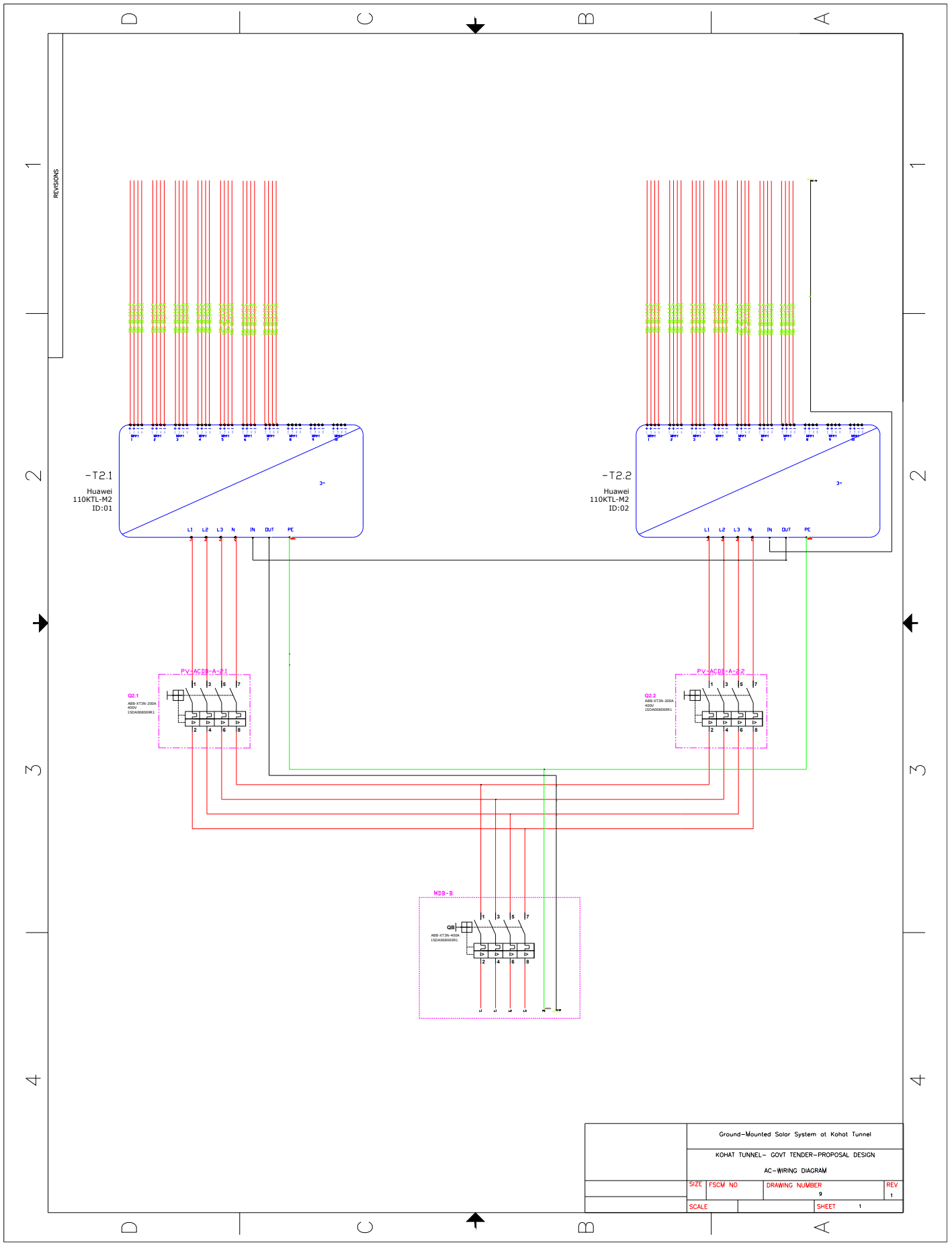

10 AWG Copper
STRING COUNT: 7; MODULE COUNT: 126
JKM550N-72HL4-BDV
DISCONNECT BOX
SUN2000-110KTL-M2 (400V)
6 AWG Aluminum
STRING COUNT: 6; MODULE COUNT: 108
STRING COUNT: 5; MODULE COUNT: 90
MASTER HIGH VOLTAGE BMS HV-BMS CONTROL PANEL
Battery Pack 1
Battery Pack 2
Battery Pack 3
COM
INVEREX LiFePO4 Cabinet Enclosure Container System
HUAWEI LUNA2000-315KTL PCS (300kW)
AC DISCONNECT
METER
GRID
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
SINGLE LINE DIAGRAM
SIZE
FSCM NO
DRAWING NUMBER
11
REV
1
SCALE
SHEET
1

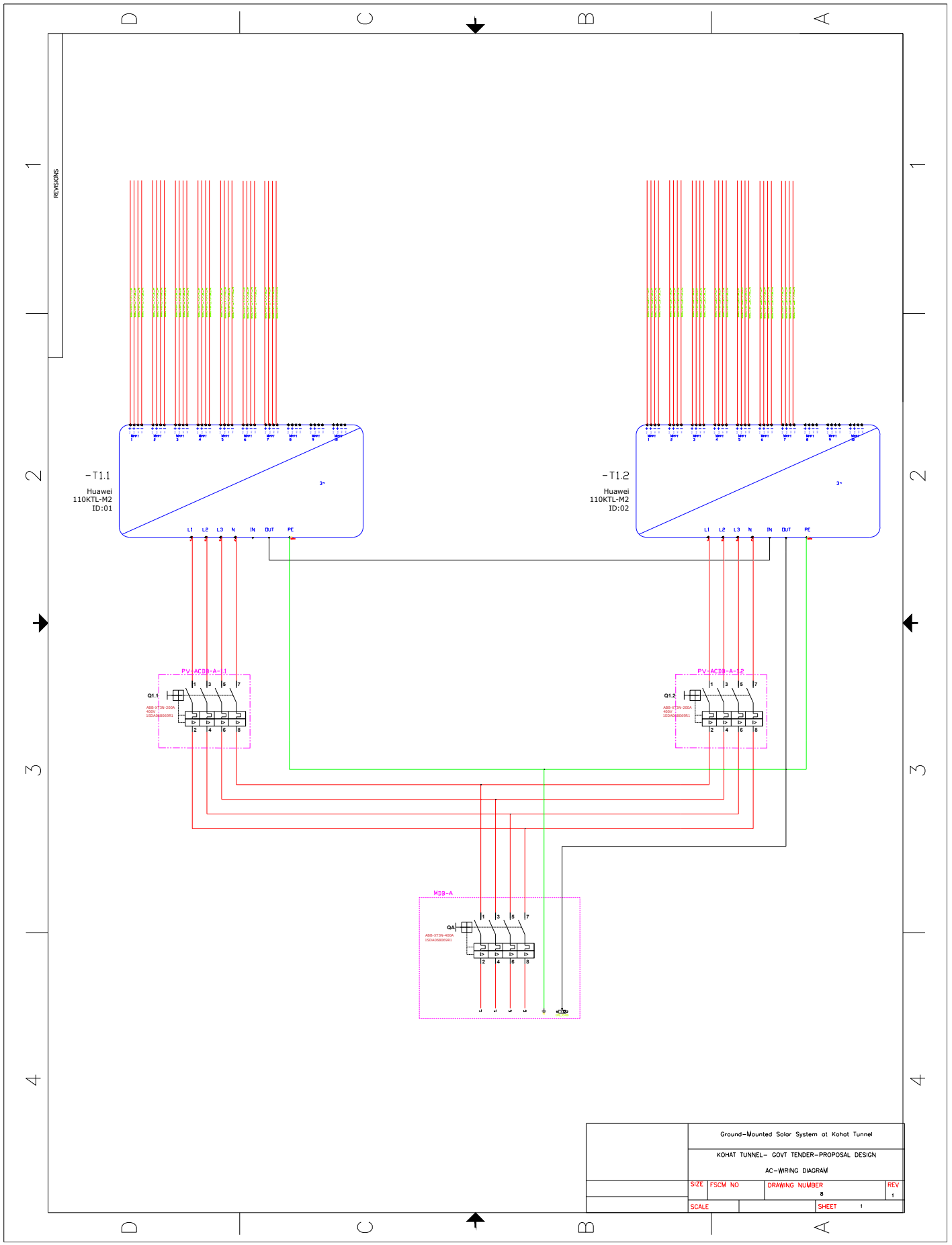

MDB-BESS
MDB-D
MDB-A
MDB-B
MDB-C
LV-BUSBAR-0.415KV
COM-BOX
kWh
SMART ENERGY METER
MAIN-LV-SWITCHES
MAIN LV SWITCHBOARD
TRANSFORMER
11kV MV Switchgear
PE
MV-BUSBAR-11KV
REVISIONS
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
AC DISTRIBUTION AND GRID CONNECTION
SIZE
FSCM NO
DRAWING NUMBER
2
REV
1
SCALE
SHEET
1


REVISIONS
DC-IN
Master DC Busbar (+)
Master DC Busbar (-)
2-Pole DC Contactor (1000V DC / 150A)
2-Pole DC Contactor (1000V DC / 150A)
2-Pole DC Contactor (1000V DC / 150A)
BATTERY PACK 01
15x Series Modules (LiFeP04)
BATTERY PACK 02
15x Series Modules (LiFeP04)
BATTERY PACK 03
15x Series Modules (LiFeP04)
CAN BUS
Sensor
MASTER HIGH VOLTAGE BMS
HV-BMS CONTROL PANEL
COM
INVEREX LiFeP04 Cabinet Enclosure Container System
DC-DISCONNECT
DC-IN
DC
AC~
COM
L1 L2 L3 N IN OUT PE
-T4.2
HUAWEI
LUNA2000-315KTL
PCS
(300kW)
MDB-BESS
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
BESS- BACKUP LAYOUT
SIZE
FSCM NO
DRAWING NUMBER
2
REV
1
SCALE
SHEET
1

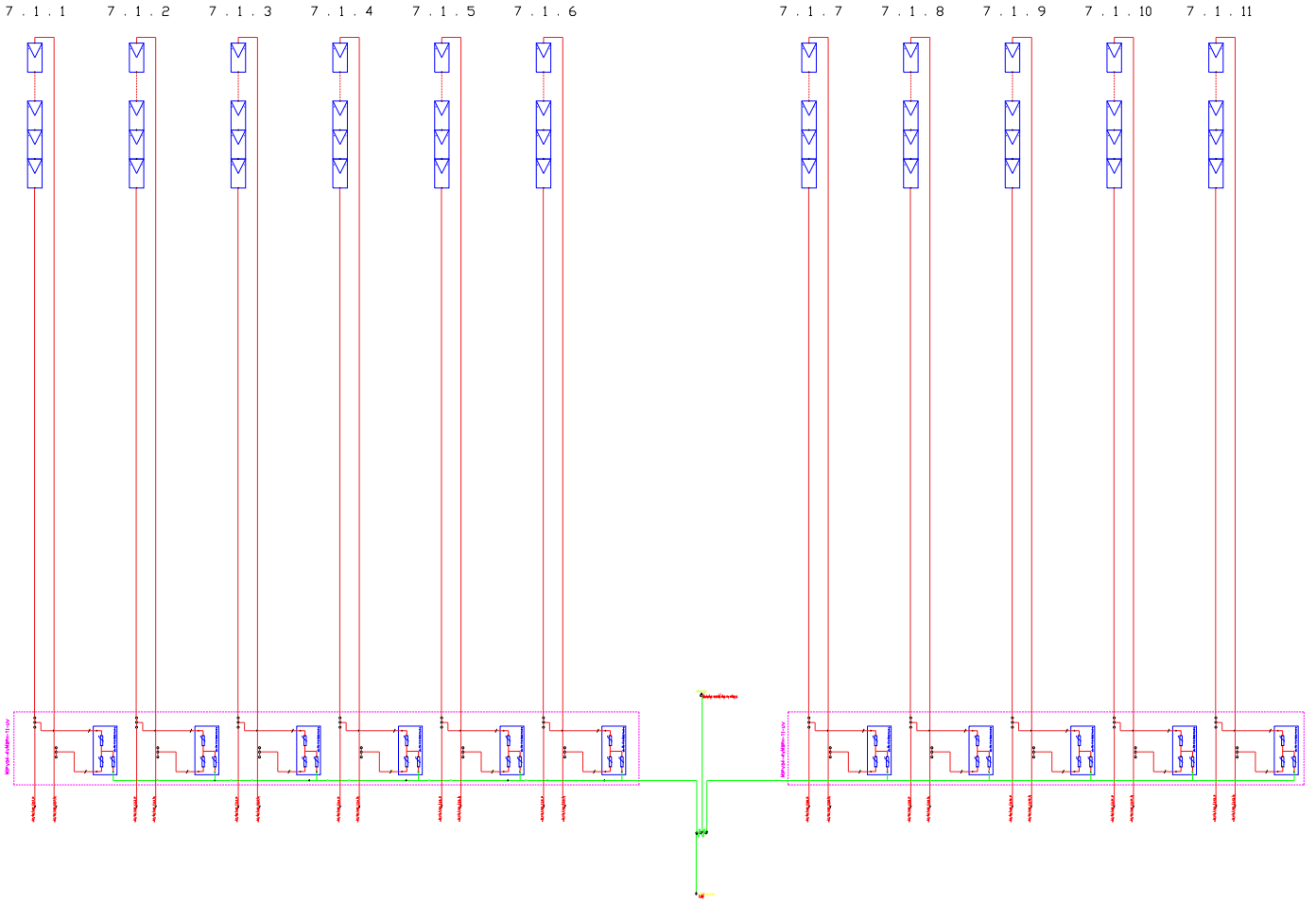

REVISIONS
-T4.1
Huawei
110KTL-M2
ID:04
3~
L1 L2 L3 N IN OUT PE
PV-ACDB-D
Q4.1
ABB-XT3N-200A
400V
1SDA068069R1
1 3 5 7
2 4 6 8
MDB-D
QD
ABB-XT3N-200A
1SDA068069R1
1 3 5 7
2 4 6 8
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
AC-WIRING DIAGRAM
SIZE
FSCM NO
DRAWING NUMBER
REV
SCALE
SHEET


REVISIONS
-T3.1
Huawei
110KTL-M2
ID:01
-T3.2
Huawei
110KTL-M2
ID:02
L1 L2 L3 N IN OUT PE
PV-ACDB-A-01
PV-ACDB-A-02
Q3.1
Q3.2
MDB-C
QC
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
AC-WIRING DIAGRAM
SIZE
FSCM NO
DRAWING NUMBER
10
REV
1
SCALE
SHEET
1


REVISIONS
-T2.1
Huawei
110KTL-M2
ID:01
-T2.2
Huawei
110KTL-M2
ID:02
L1 L2 L3 N IN OUT PE
PV-ACDB-A-2.1
PV-ACDB-A-2.2
Q2.1
Q2.2
ABB-XT3N-200A
400V
1SDA068069R1
MDB-B
QB
ABB-XT3N-400A
1SDA068056R1
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
AC-WIRING DIAGRAM
SIZE
FSCM NO
DRAWING NUMBER
9
REV
1
SCALE
SHEET
1

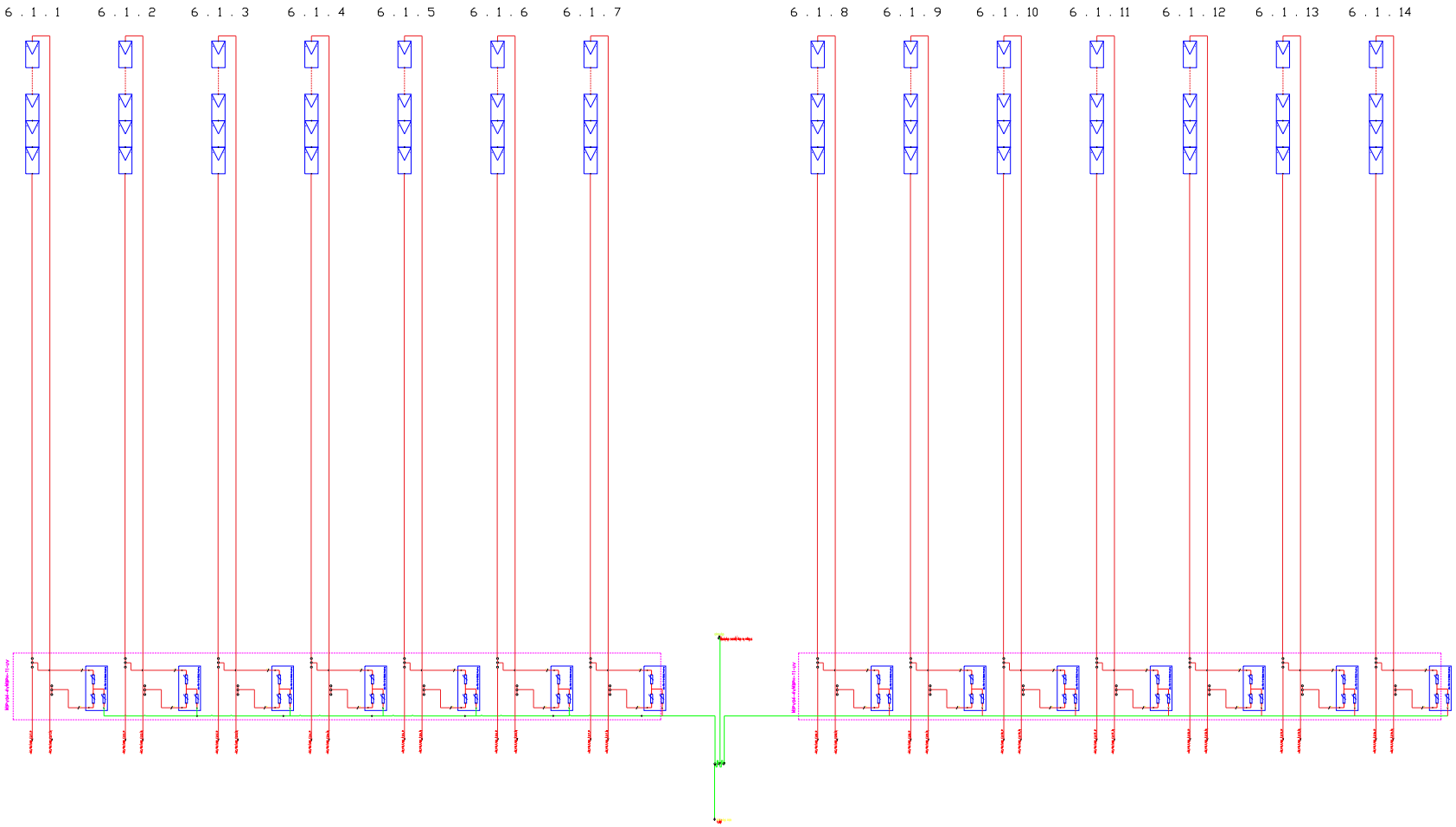

REVISIONS
-T1.1
Huawei
110KTL-M2
ID:01
-T1.2
Huawei
110KTL-M2
ID:02
L1 L2 L3 N IN OUT PE
PV-ACDB-A-1
PV-ACDB-A-2
Q1.1
Q1.2
MDB-A
QA
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
AC-WIRING DIAGRAM
SIZE
FSCM NO
DRAWING NUMBER
8
REV
1
SCALE
SHEET
1

REVISIONS



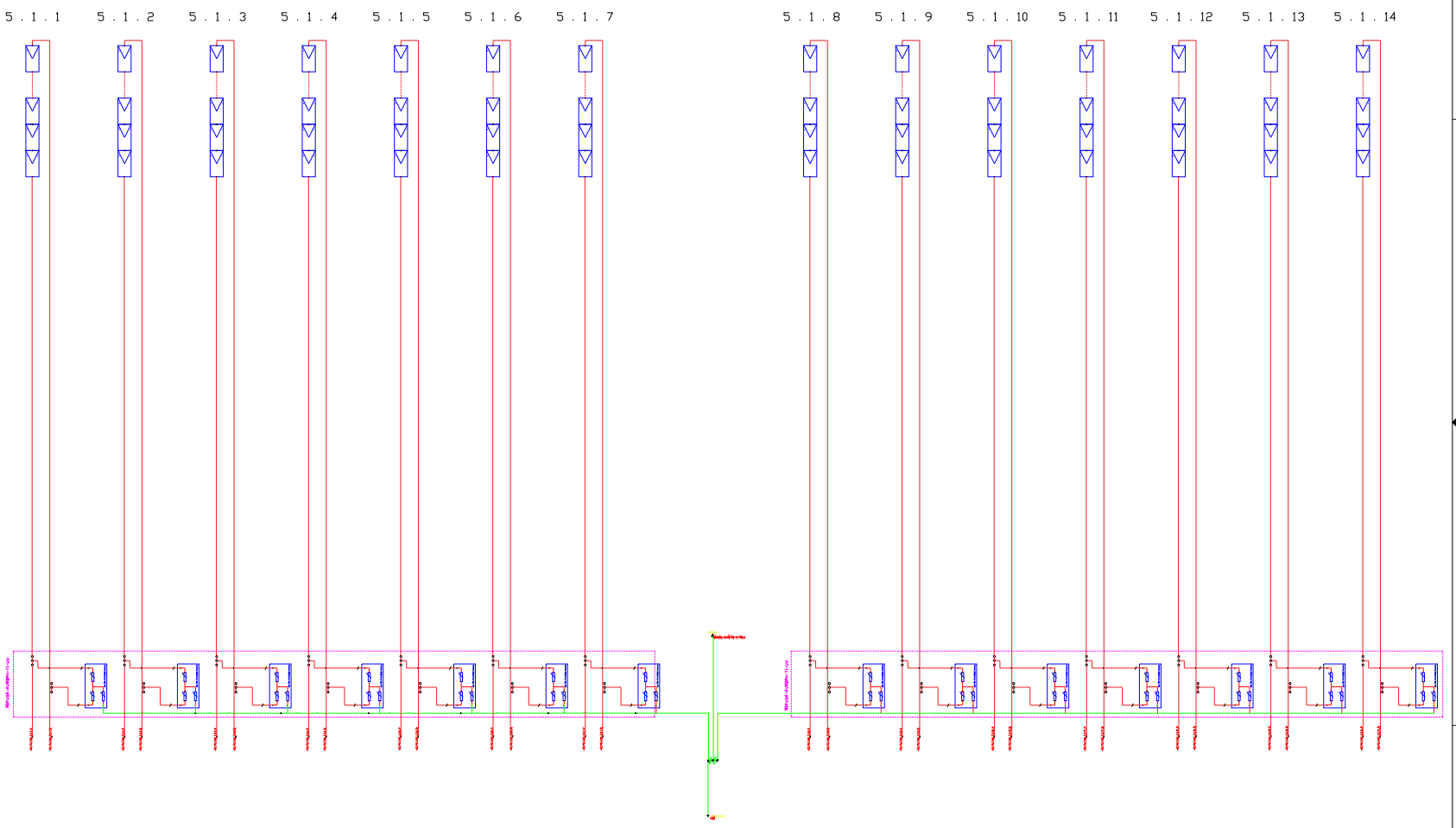
	Ground–Mounted Solar System at Kohat Tunnel
	KOHAT TUNNEL– GOVT TENDER–PROPOSAL DESIGN
	PV Array 7
SIZE	FSCM NO / DRAWING NUMBER: 7 / REV: 1
SCALE	SHEET: 1

REVISIONS



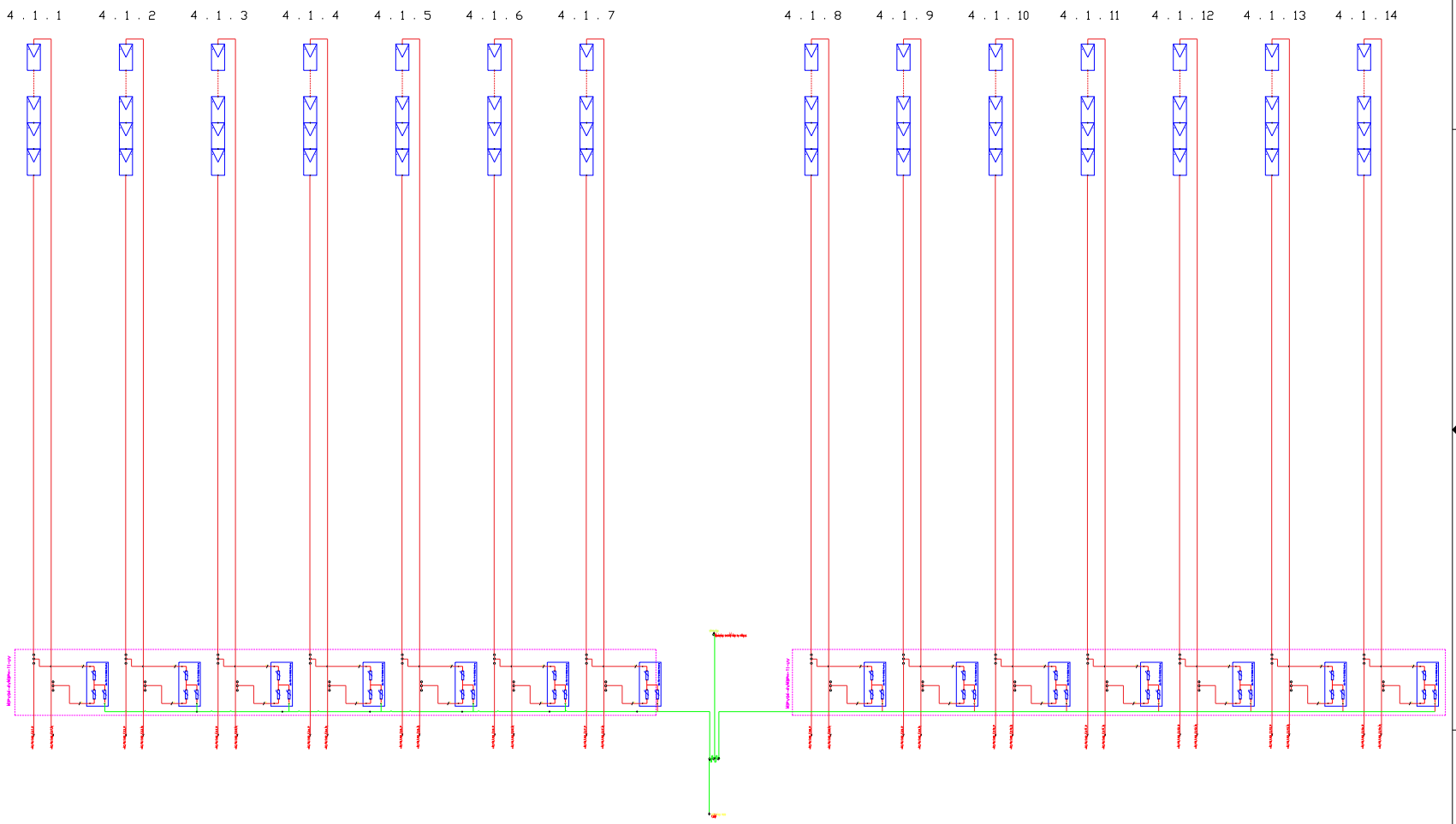
	Ground-Mounted Solar System at Kohat Tunnel
	KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
	PV Array 6
SIZE / FSCM NO / DRAWING NUMBER / REV	6 / 1
SCALE / SHEET	1

REVISIONS			



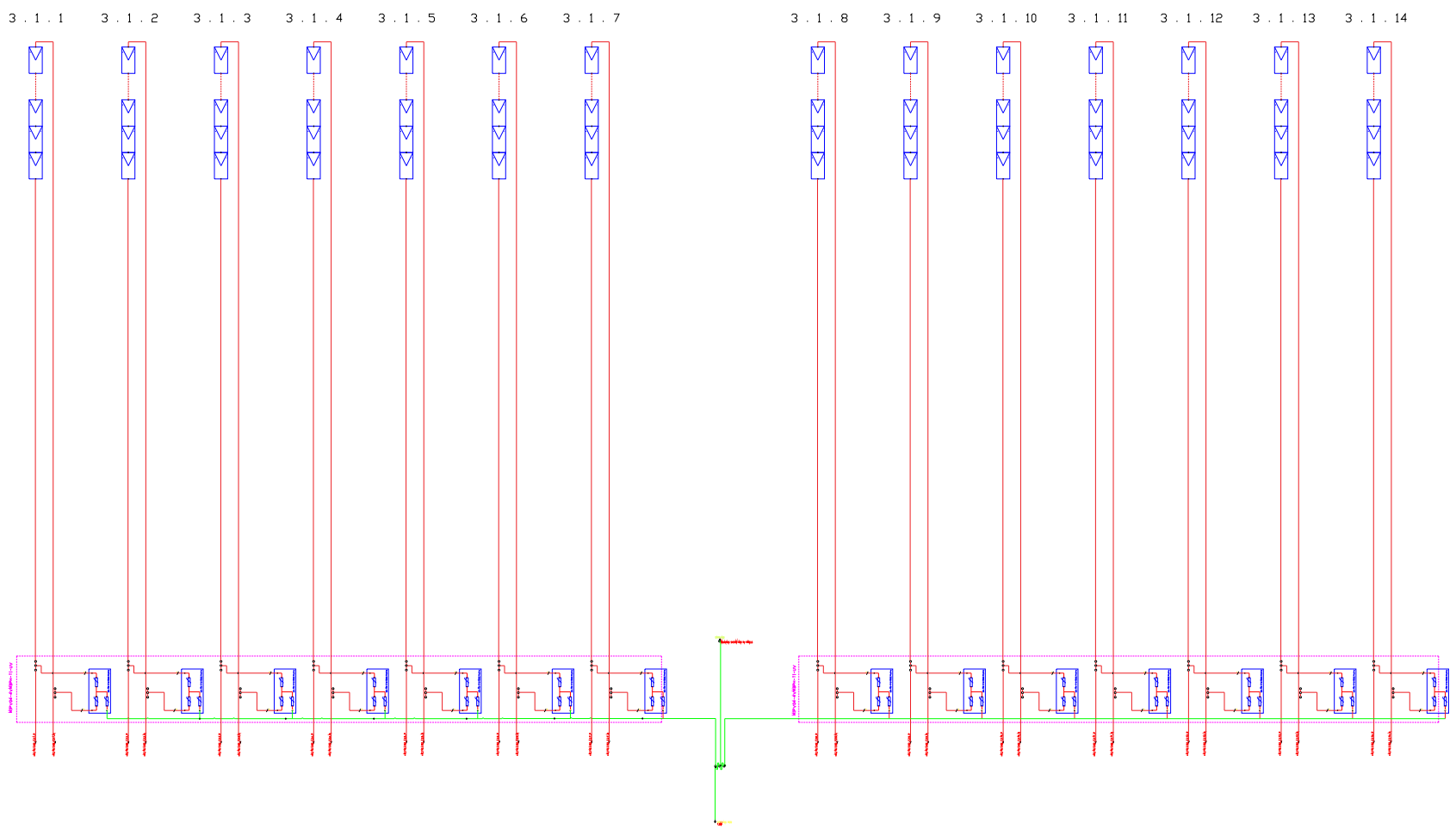
	Ground-Mounted Solar System at Kohat Tunnel			
	KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN			
	PV Array 5			
	SIZE	FSCM NO	DRAWING NUMBER 5	REV 1
	SCALE		SHEET 1	

REVISIONS					



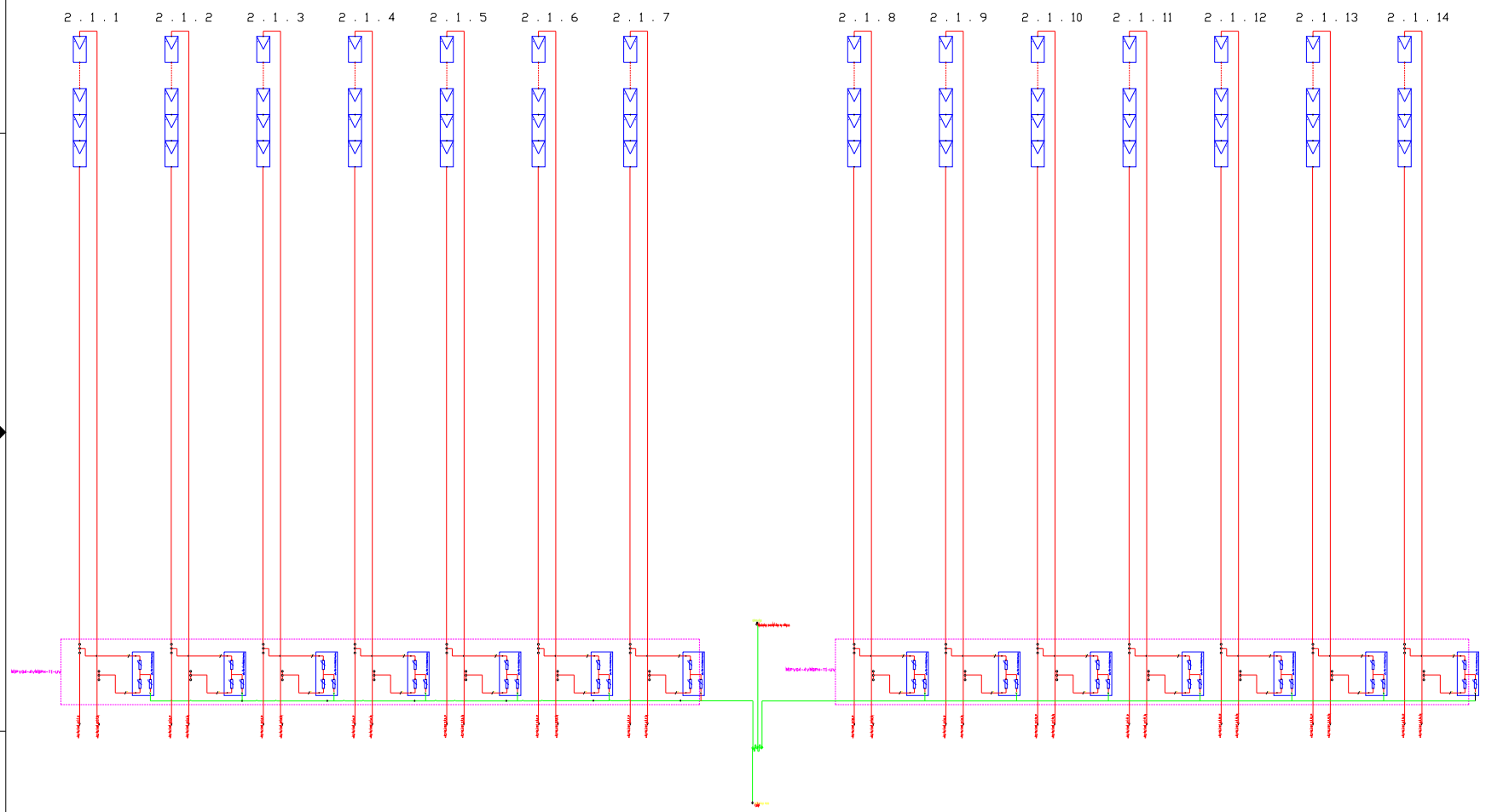
	Ground-Mounted Solar System at Kohat Tunnel			
	KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN			
	PV Array 4			
	SIZE	FSCM NO	DRAWING NUMBER 4	REV 1
	SCALE		SHEET 1	

REVISIONS					

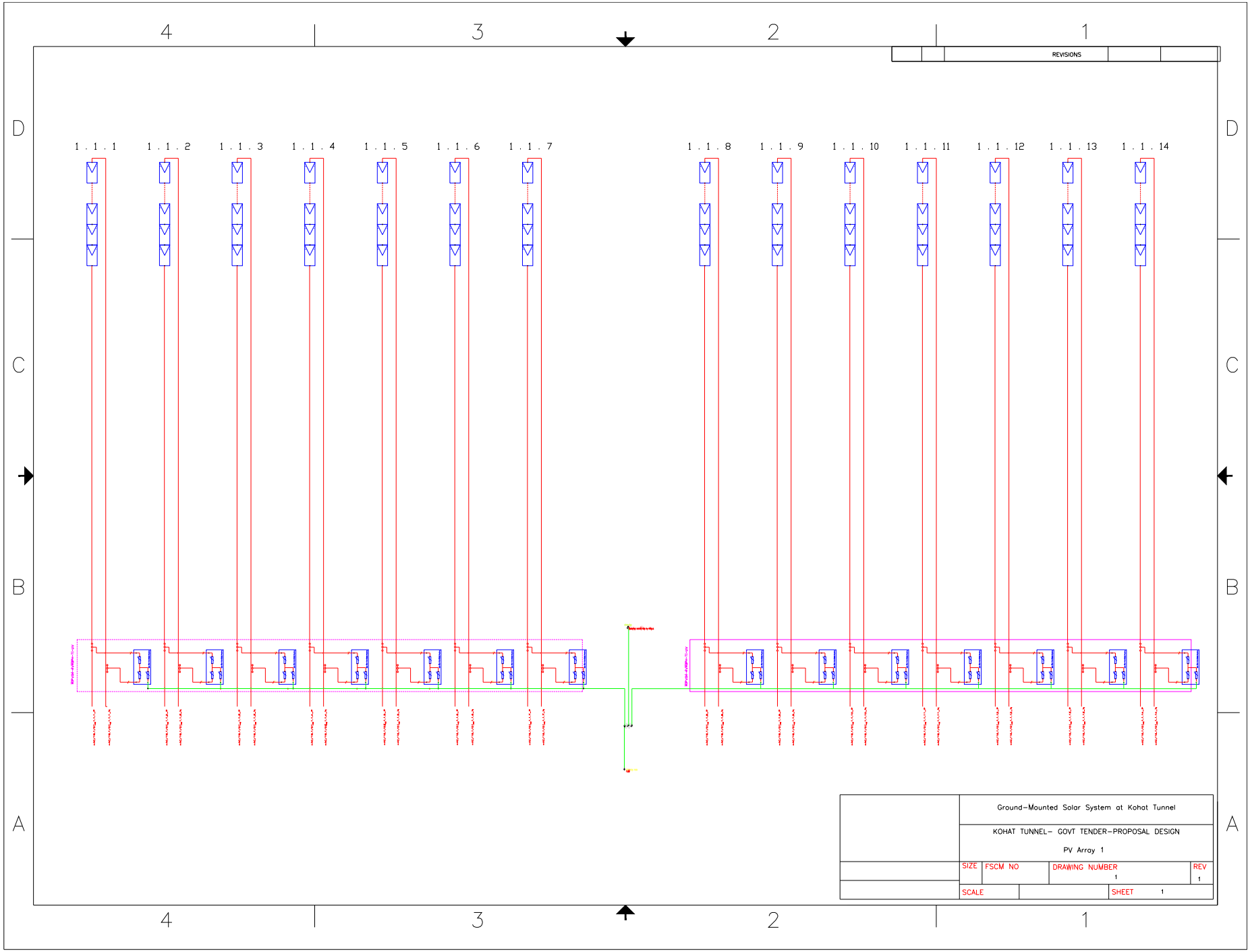


	Ground-Mounted Solar System at Kohat Tunnel			
	KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN			
	PV Array 3			
	SIZE	FSCM NO	DRAWING NUMBER 3	REV 1
	SCALE		SHEET 1	

REVISIONS				



	Ground-Mounted Solar System at Kohat Tunnel
	KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
	PV Array 2
SIZE / FSCM NO / DRAWING NUMBER / REV	2 / 1
SCALE / SHEET	1


4
3
2
1
D
C
B
A
REVISIONS
1 . 1 . 1
1 . 1 . 2
1 . 1 . 3
1 . 1 . 4
1 . 1 . 5
1 . 1 . 6
1 . 1 . 7
1 . 1 . 8
1 . 1 . 9
1 . 1 . 10
1 . 1 . 11
1 . 1 . 12
1 . 1 . 13
1 . 1 . 14
Ground-Mounted Solar System at Kohat Tunnel
KOHAT TUNNEL- GOVT TENDER-PROPOSAL DESIGN
PV Array 1
SIZE
FSCM NO
DRAWING NUMBER
1
REV
1
SCALE
SHEET
1